

TF-blood_vs_primary

2026-06-01

Setup and data loading

```
library(tidyverse)
library(here)
library(corrplot)
data <- readxl::read_xlsx(here('data', 'processed', 'PCAWG_primary.xlsx'))
clean_data <- data
summary(clean_data)
```

donor_id	cancer_type	alt_status	tf_primary_rate
Length:2014	Length:2014	Length:2014	Min. : 0.00000
Class :character	Class :character	Class :character	1st Qu.: 0.00000
Mode :character	Mode :character	Mode :character	Median : 0.03931
			Mean : 0.13568
			3rd Qu.: 0.10251
			Max. :19.73776

tf_blood_rate	telomere_insertion_rate	telomere_content_log2
Min. :0.00000	Min. :0.00000	Min. : -4.00250
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.: -0.92876
Median :0.02628	Median :0.00000	Median : -0.52968
Mean :0.04660	Mean :0.03307	Mean : -0.43589
3rd Qu.:0.05967	3rd Qu.:0.02195	3rd Qu.: -0.07054
Max. :1.53097	Max. :5.38979	Max. : 3.84469

TERT_FPKM	ATAGGG_singleton_dist	CTAGGG_singleton_dist
Min. : 0.00000	Min. : -4.35757	Min. : -3.49674
1st Qu.: 0.02069	1st Qu.: -0.68156	1st Qu.: -0.38418

Median : 0.10630	Median :-0.03350	Median : 0.05423
Mean : 0.90171	Mean :-0.06042	Mean : 0.06588
3rd Qu.: 0.37979	3rd Qu.: 0.57645	3rd Qu.: 0.53662
Max. :53.76790	Max. : 4.47470	Max. : 7.80772
NA's :1166	NA's :28	NA's :2

GTAGGG_singleton_dist	TAAGGG_singleton_dist	TCAGGG_singleton_dist
Min. :-2.981384	Min. :-4.67195	Min. :-4.27208
1st Qu.: -0.359368	1st Qu.: -0.53849	1st Qu.: -0.56813
Median : 0.017444	Median : 0.08092	Median : 0.02901
Mean : 0.001582	Mean : 0.06257	Mean : 0.08148
3rd Qu.: 0.352901	3rd Qu.: 0.67706	3rd Qu.: 0.70046
Max. : 4.253404	Max. : 3.87943	Max. : 7.64722
	NA's :28	NA's :18

TGAGGG_singleton_dist	TTCGGG_singleton_dist	TTGGGG_singleton_dist
Min. :-5.27154	Min. :-3.848796	Min. :-3.109601
1st Qu.: -0.37005	1st Qu.: -0.568133	1st Qu.: -0.417046
Median : 0.06673	Median : 0.007297	Median : -0.019229
Mean : 0.07103	Mean : 0.102063	Mean : 0.002303
3rd Qu.: 0.46860	3rd Qu.: 0.573244	3rd Qu.: 0.429475
Max. : 6.67267	Max. : 8.739318	Max. : 4.584889
	NA's :5	NA's :2

TTTGGG_singleton_dist	Specimen.Type.Summary	cancer_group
Min. :-4.1941	Length:2014	Length:2014
1st Qu.: -0.5610	Class :character	Class :character
Median : -0.1030	Mode :character	Mode :character
Mean :-0.1309		
3rd Qu.: 0.3465		
Max. : 2.7484		

Distributions

Distributions - summary

```
summary(clean_data$tf_blood_rate)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.02628	0.04660	0.05967	1.53097

```
summary(clean_data$tf_primary_rate)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.00000	0.03931	0.13568	0.10251	19.73776

```
zero_blood <- sum(clean_data$tf_blood_rate == 0)/nrow(clean_data) * 100  
zero_primary <- sum(clean_data$tf_primary_rate == 0)/nrow(clean_data) * 100  
  
cat("Percentuale di zeri in tf_blood_rate: ", zero_blood, "%\n")
```

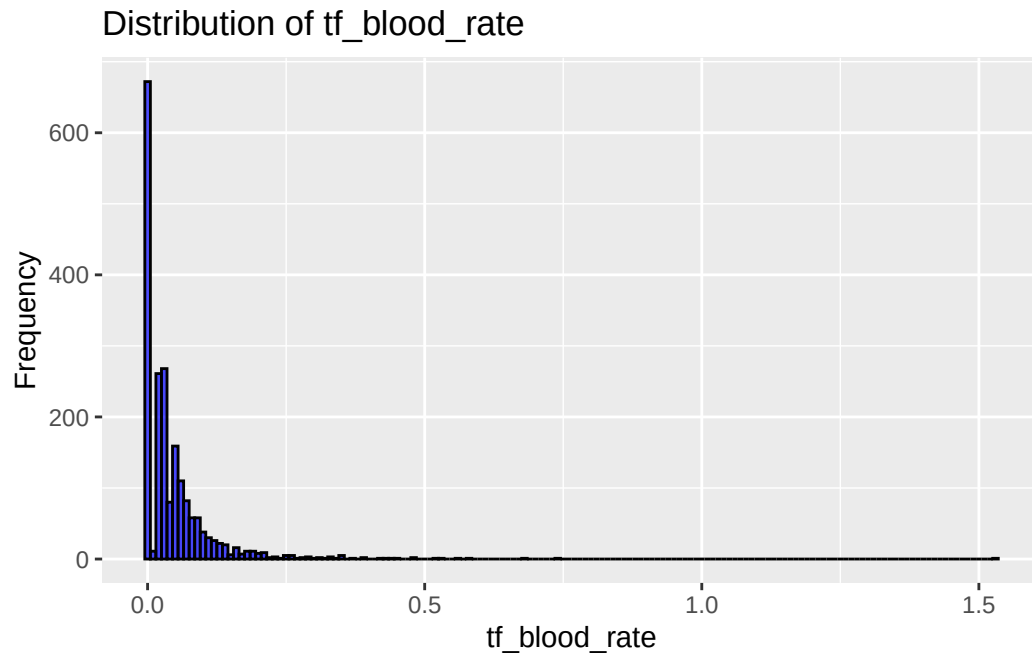
Percentuale di zeri in tf_blood_rate: 33.36643 %

```
cat("Percentuale di zeri in tf_primary_rate: ", zero_primary, "%\n")
```

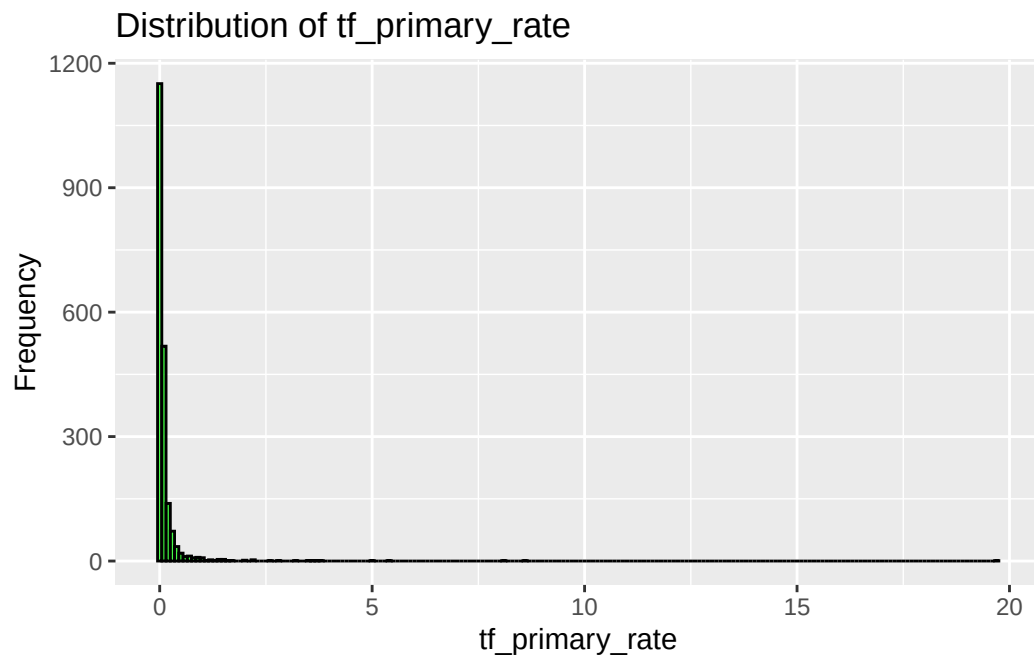
Percentuale di zeri in tf_primary_rate: 27.95432 %

Distributions - visual

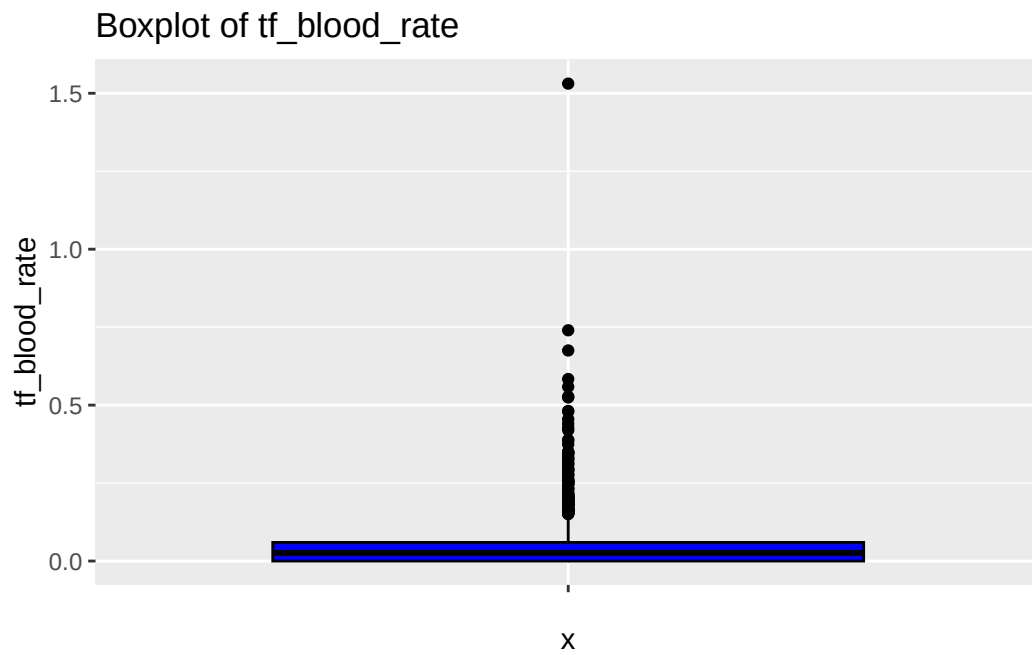
```
clean_data %>%  
  ggplot(aes(x = tf_blood_rate)) +  
  geom_histogram(binwidth = 0.01, fill = "blue", color = "black", alpha = 0.7) +  
  labs(title = "Distribution of tf_blood_rate", x = "tf_blood_rate", y = "Frequency")
```



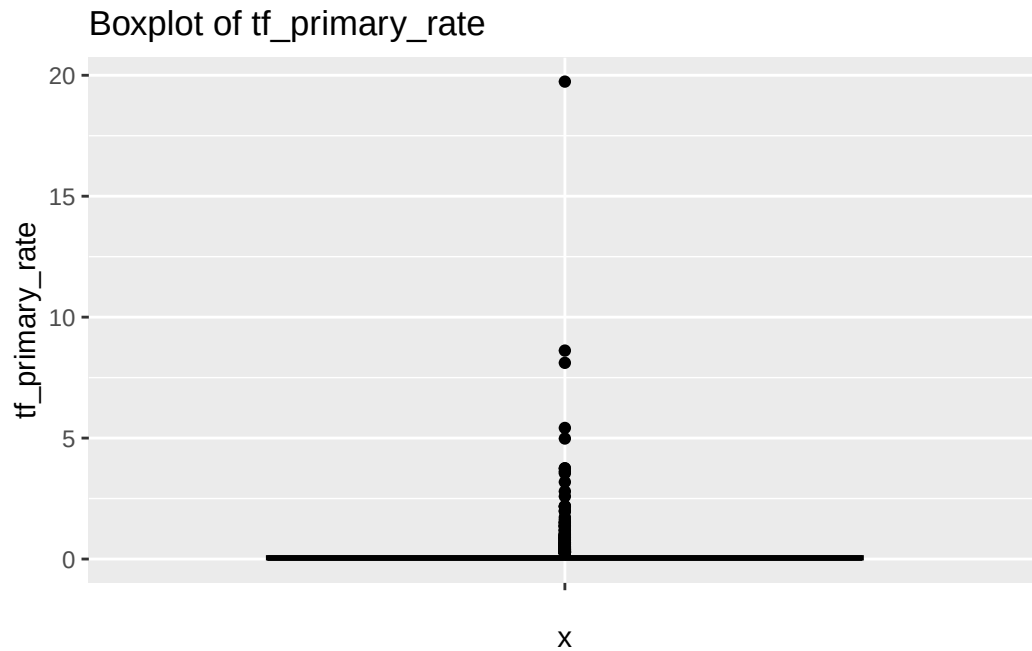
```
clean_data %>%  
  ggplot(aes(x = tf_primary_rate)) +  
  geom_histogram(binwidth = 0.1, fill = "green", color = "black", alpha = 0.7) +  
  labs(title = "Distribution of tf_primary_rate", x = "tf_primary_rate", y = "Frequency")
```



```
clean_data %>%  
  ggplot(aes(x = "", y = tf_blood_rate)) +  
  geom_boxplot(fill = "blue", color = "black") +  
  labs(title = "Boxplot of tf_blood_rate", y = "tf_blood_rate")
```



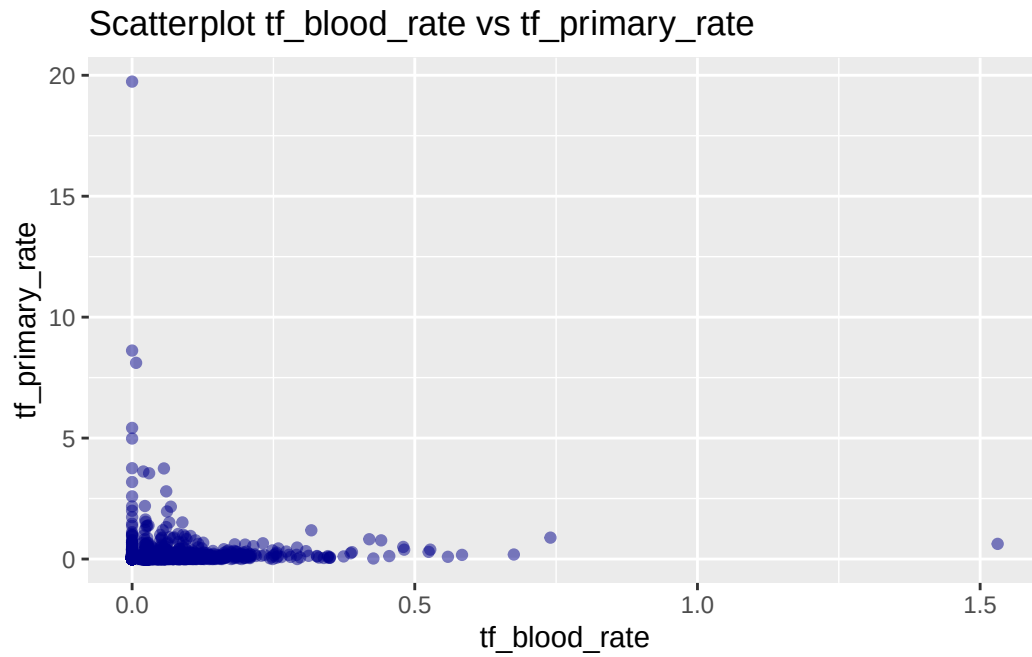
```
clean_data %>%  
  ggplot(aes(x = "", y = tf_primary_rate)) +  
  geom_boxplot(fill = "green", color = "black") +  
  labs(title = "Boxplot of tf_primary_rate", y = "tf_primary_rate")
```



Scatter plots

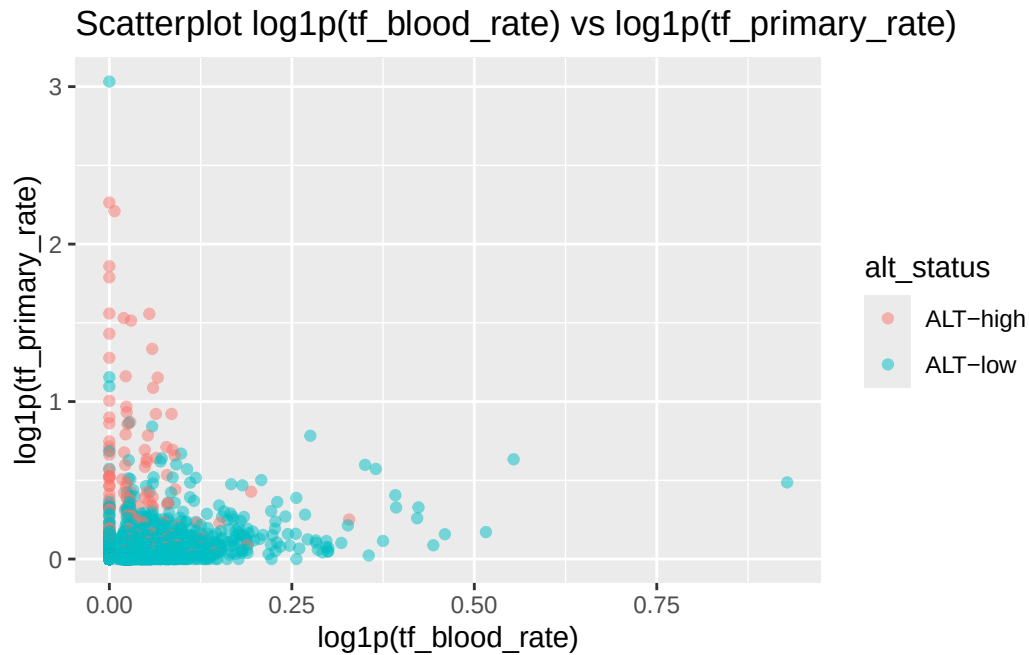
Scatterplot raw

```
clean_data %>%  
  ggplot(aes(tf_blood_rate, tf_primary_rate)) +  
  geom_point(alpha = 0.5, color = "darkblue") +  
  labs(title = "Scatterplot tf_blood_rate vs tf_primary_rate", x = "tf_blood_rate", y = "tf_
```



Scatterplot log1p transformed

```
clean_data %>%  
  ggplot(aes(log1p(tf_blood_rate), log1p(tf_primary_rate), color = alt_status)) +  
  geom_point(alpha = 0.5) +  
  labs(title = "Scatterplot log1p(tf_blood_rate) vs log1p(tf_primary_rate)", x = "log1p(tf_b
```

Distinction zero vs positive

```
clean_data <- clean_data %>%
  mutate(blood_pos = ifelse(tf_blood_rate > 0, 1, 0),
         primary_pos = ifelse(tf_primary_rate > 0, 1, 0))

table(clean_data$blood_pos, clean_data$primary_pos, dnn = c('blood_pos', 'primary_pos'))
```

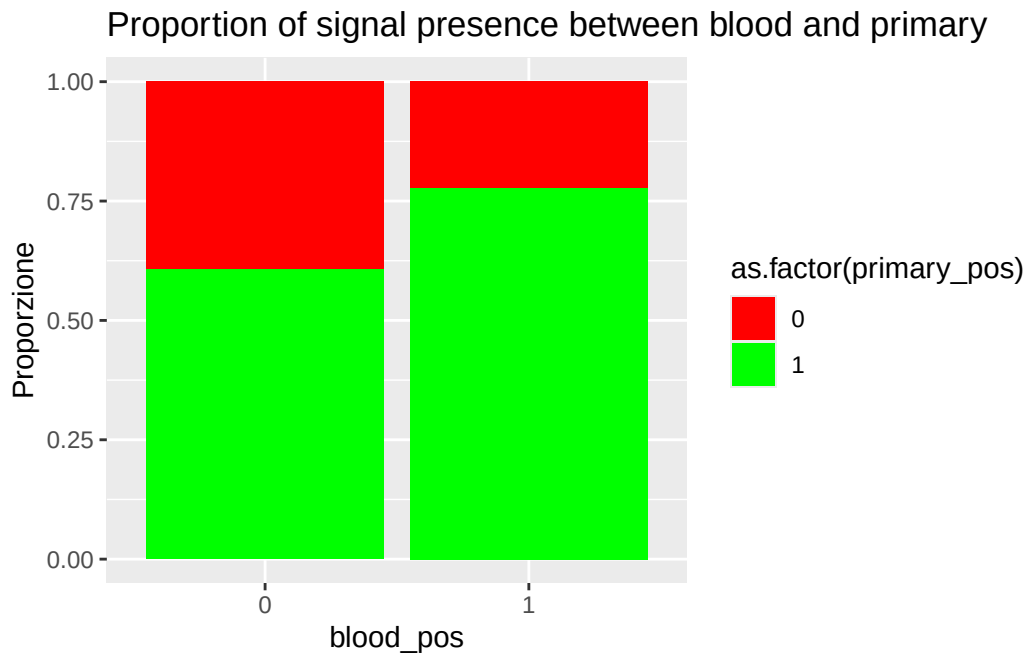
	primary_pos	
blood_pos	0	1
0	264	408
1	299	1043

```
chisq.test(table(clean_data$blood_pos, clean_data$primary_pos))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(clean_data$blood_pos, clean_data$primary_pos)
X-squared = 63.455, df = 1, p-value = 1.641e-15
```

```
ggplot(clean_data, aes(x = as.factor(blood_pos), fill = as.factor(primary_pos))) +
  geom_bar(position = "fill") +
  labs(title = "Proportion of signal presence between blood and primary", x = "blood_pos", y = "Proportion") +
  scale_fill_manual(values = c("red", "green"), labels = c("0", "1"))
```



Discordant samples: primary TF = 0, blood TF > 0

Definition and ALT status enrichment

```
# Samples with no detectable primary TF but positive blood TF
discordant <- clean_data %>%
  mutate(group = case_when(
    primary_pos == 0 & blood_pos == 1 ~ "primary=0, blood>0",
    primary_pos == 1 & blood_pos == 1 ~ "both>0",
    primary_pos == 1 & blood_pos == 0 ~ "primary>0, blood=0",
    TRUE ~ "both=0"
  ))

cat("Group counts:\n")
```

Group counts:

```
print(table(discordant$group))
```

both=0	both>0 primary=0, blood>0	primary>0, blood=0
264	1043	299
		408

```
# Focus on discordant group vs the rest (excluding both=0 as uninformative)
disc_subset <- discordant %>%
  filter(group %in% c("primary=0, blood>0", "both>0", "primary>0, blood=0")) %>%
  mutate(is_discordant = group == "primary=0, blood>0")

cat("\nALT status in discordant group (primary=0, blood>0):\n")
```

ALT status in discordant group (primary=0, blood>0):

```
print(table(discordant$alt_status[discordant$group == "primary=0, blood>0"]))
```

ALT-high	ALT-low
9	290

```
cat("\nALT status in non-discordant groups:\n")
```

ALT status in non-discordant groups:

```
print(table(discordant$alt_status[discordant$group != "primary=0, blood>0"]))
```

ALT-high	ALT-low
161	1554

```
# Fisher's exact test: is ALT-low enriched in discordant samples?
ct <- table(disc_subset$is_discordant, disc_subset$alt_status)
rownames(ct) <- c("other", "discordant")
print(ct)
```

	ALT-high	ALT-low
other	150	1301
discordant	9	290

```
fisher_res <- fisher.test(ct)
cat("\nFisher's exact test (discordant vs other groups, ALT-low enrichment):\n")
```

Fisher's exact test (discordant vs other groups, ALT-low enrichment):

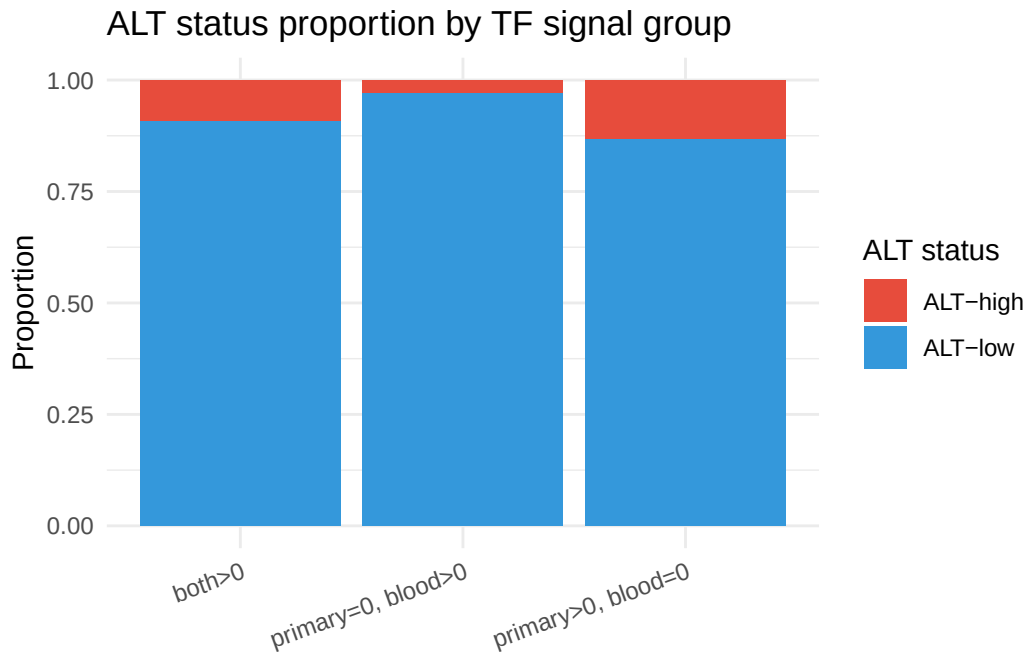
```
print(fisher_res)
```

Fisher's Exact Test for Count Data

```
data: ct
p-value = 1.261e-05
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.874170 8.376733
sample estimates:
odds ratio
 3.713292
```

ALT status proportion plot

```
# Bar chart: ALT status proportions by group
discordant %>%
  filter(group != "both=0") %>%
  ggplot(aes(x = group, fill = alt_status)) +
  geom_bar(position = "fill") +
  scale_fill_manual(values = c("ALT-high" = "#E74C3C", "ALT-low" = "#3498DB")) +
  labs(
    title = "ALT status proportion by TF signal group",
    x = NULL, y = "Proportion", fill = "ALT status"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))
```



Discordant rate by cancer type

```
disc_pct <- discordant %>%
  group_by(cancer_type) %>%
  summarise(
    n_total = n(),
    n_disc = sum(group == "primary=0, blood>0"),
    pct_disc = n_disc / n_total * 100,
    .groups = "drop"
  ) %>%
  filter(n_total >= 10) %>%
  arrange(desc(pct_disc))

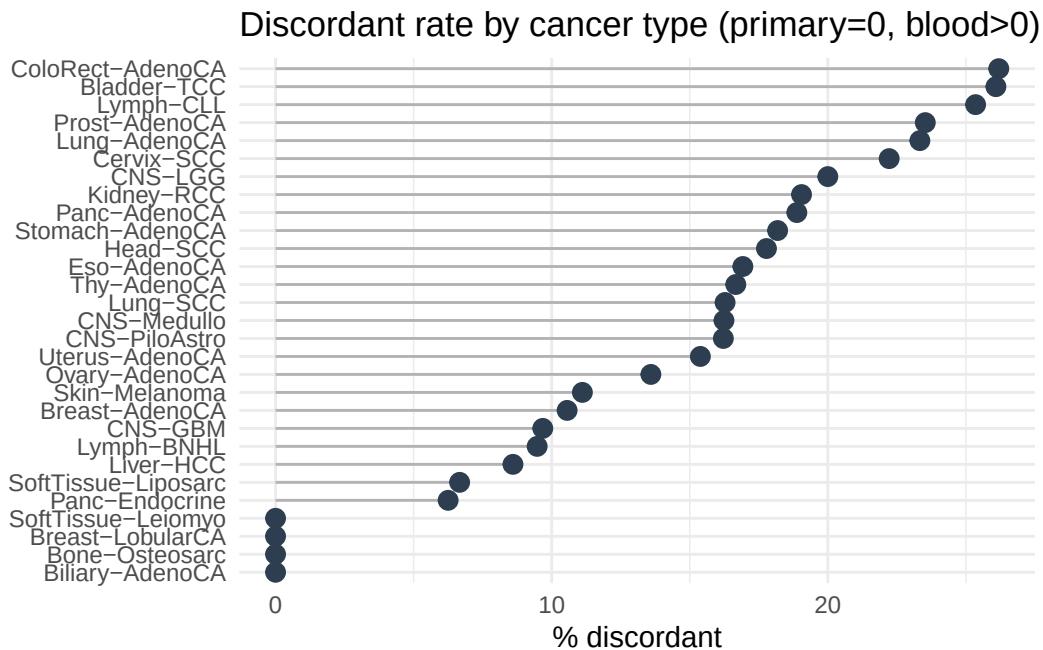
print(disc_pct)
```

```
# A tibble: 29 x 4
  cancer_type      n_total n_disc pct_disc
  <chr>          <int>  <int>   <dbl>
1 ColoRect-AdenoCA    42     11    26.2
2 Bladder-TCC        23      6    26.1
3 Lymph-CLL          71     18    25.4
```

4 Prost-AdenoCA	136	32	23.5
5 Lung-AdenoCA	30	7	23.3
6 Cervix-SCC	18	4	22.2
7 CNS-LGG	15	3	20
8 Kidney-RCC	126	24	19.0
9 Panc-AdenoCA	196	37	18.9
10 Stomach-AdenoCA	55	10	18.2

i 19 more rows

```
ggplot(disc_pct, aes(x = pct_disc, y = reorder(cancer_type, pct_disc))) +
  geom_segment(aes(x = 0, xend = pct_disc,
                  y = reorder(cancer_type, pct_disc),
                  yend = reorder(cancer_type, pct_disc)),
              color = "grey70") +
  geom_point(size = 3, color = "#2C3E50") +
  labs(
    title = "Discordant rate by cancer type (primary=0, blood>0)",
    x = "% discordant", y = NULL
  ) +
  theme_minimal()
```



Spearman correlation

```
positive_data <- clean_data %>%  
  filter(blood_pos == 1 & primary_pos == 1)  
  
cor.test(positive_data$tf_blood_rate[positive_data$blood_pos == 1],  
         positive_data$tf_primary_rate[positive_data$primary_pos == 1],  
         method = 'spearman')
```

Spearman's rank correlation rho

```
data: positive_data$tf_blood_rate[positive_data$blood_pos == 1] and positive_data$tf_primary_rate[positive_data$primary_pos == 1]  
S = 144483840, p-value = 1.401e-14  
alternative hypothesis: true rho is not equal to 0  
sample estimates:  
      rho  
0.2359567
```

Relationship visualization

```
ggplot(positive_data, aes(x = log1p(tf_blood_rate), y = log1p(tf_primary_rate))) +  
  geom_point(alpha = 0.5, color = "darkred") +  
  geom_smooth(method = "loess", color = "blue") +  
  labs(title = "tf_blood_rate vs tf_primary_rate", x = "log1p(tf_blood_rate)", y = "log1p(tf_primary_rate)")
```

